

# ActInf Livestream #037.0 ~ "Free Energy: A User's Guide"

Livestream | 1:29:27

Thank you.

All right, it's January 28th, 2022, and we're here in Actonflab livestream number 37.0, discussing the paper Free Energy, a User's Guide.

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All right, so today in 37.0, we are going to learn and discuss and introduce and provide some background for this paper, Free Energy, a User's Guide, by Stephen Francis Mann, Ross Payne, and Michael Kirchhoff.

And just like all the dot zeros and discussions, we're just going to introduce some ideas, give a little tip of the iceberg, and kind of warm up to talk about this paper in the coming two weeks.

And so today we'll be going through a lot of the background and some of the key formalisms in the paper.

And in the coming two weeks, we'll be discussing the paper, so let us know if you want to join.

So first, there's this comment in red.

Is this yours, Dean?

Yeah.

Do you want to go over here?

Pardon me?

Do you want to describe it here or in your intro?

Yeah, in the intro.

Great.

All right.

Well, we'll each just go around and say hello.

And then I think we can just say one reason why we're excited to discuss it.

I'm Daniel.

I'm in California and was just enticed by the title, short and direct.

And I think it has some great solidifications and perspectives and implications on multiple levels.

So how about Stephen?

Hello.

I'm Stephen.

I'm based in Toronto.

Yeah, I was interested by another sort of way to surmise the free energy principle and active inference type approaches.

in relation to the philosophy of biology and cognitive science.

I thought that was interesting to see how that pitch was being made.

Over to Dean.

Hi, I'm Dean.

I'm in Calgary.

And anytime I see the word guide, I'm always curious because of my background in setting up programming where people were going into novel situations.

So the authors kind of had me at that word.

And then I wanted to sort of learn more about what they had to say.

What are some key features of guides or things that guides should keep in mind before we head off on this journey?

Yeah, well, for me, I sort of adopted the term wayfinding.

Kind of as a blanket, not just a mark of a blanket, but as a way of maybe describing what that path could turn out to be without necessarily constraining people.

So, and there, I'm sure there are other descriptions and definitions of the word guide, but that's the one I tend to, to, uh, gravitate towards.

All right.

Steven.

Yeah.

And there's also what kind of guide we end up inhabiting.

So, you know, this biological grounding, it feels like, uh, there's a desire to get into our feet and into the soil somehow, uh, as well as the cognitive science.

So I feel it's a bit like I've been stretched up and, uh, pushed down into the ground.

So there's something there.

Okay.

So one big question, probably not the only big question that could be raised, but just the question that approached us to this paper in some way is how can we learn and apply active inference?

And then Dean, do you want to talk about what you wrote here?

Well, I just like to kind of read it out.

I don't.

Yes.

Okay.

So assuming that making moves, which we could describe as behavior now has effects on what happens next and that those moves are at least.

loosely based or inferred on evident and the, and the, and the authors use the terms, probabilities, predictions, and fitness.

So that's in the paper.

Then how might we begin to look at energy applied?

Okay.

And in calculating out our best guess is measure, measurable as or of what we think or believe will happen next.

Step one, that road to prediction matter expertise.

Now prediction matter expertise is a term I coined and brought up with Dr. Friston.

Um, but I think it might, if we're guiding, we might think about that in the background.

The other part was they talk about models and blast in our last live stream set.

That's all we were doing is talk about modeling ourselves.

So it seems like a natural extension.

Now it's talking about models, models as inference and action models, as selection models, as extenders, i.e. tools.

And then how does a model curate, curated, rundown, help us get outside the guide when using energy in ways beyond guide explication and interpretation?

Do we need a user's guide for the user's guide?

Which is something that I used to ask all the time, uh, the kids that were participating in the program and that I was setting up.

Um, because they were going out with their background and they had sponsors who had a different background.

And we often wondered whether or not they would have to give a user's guide to somebody who is necessarily more experienced with the field, but not necessarily experienced with bringing young personal.

on. So that's why I think that when we're talking about how can we learn and apply

inference in this guide theme, those are things we might want to keep sort of at least partially

in our thinking. Thanks a lot for sharing that, Dean. A lot to say, but that's what the dot one

and the dot two and beyond are for. So the paper is Free Energy, a User's Guide by Stephen Francis

Mann, Ross Payne and Michael Kirchhoff. It's from The Last Days of 21, I believe, and it is

on the PhilSci Archive. Okay, so here are some of the aims and the claims of the paper.

Over the past 15 years, a novel explanatory framework spearheaded by Carl Furstén has inspired both

excitement and confusion of philosophy of biology and cognitive science. Active inference, whose

most famous tenet is the free energy principle, purports to unify explanations in biology and

cognitive science under a single class of mathematical models. So not math under biology, but biology under

math. There are broadly three reasons why the active inference framework is difficult to understand.

In the author's words, first, the mathematics are unfamiliar to many philosophers and even to

biologists and cognitive scientists and mathematicians and all other kinds of peoples. Second, the framework

was developed rapidly by a small but dedicated group of researchers limiting its accessibility while

expanding its scope. Hashtag recent history. Third, the framework makes claims across both mathematical

and empirical domains and the dialectical relationships between these are unclear. So what is the math,

bio-dialectic, bio-dialectic? Who's nested within who? Yes, Stephen.

Stephen. Yeah, I think this, the way these three kind of points are made is useful for seeing the rationale

for this paper. I think there's seen as a desire to come at these questions as if it was coming from

people less steeped in all the math. Because previous papers might have come out which had an awful lot of particular, you know, particular information that comes from this dedicated group of researchers.

I think by trying to speak about it from the perspective of people arriving, it gives those core

researchers of which I know for sure Michael Kirchhoff is, you know, been part of, a chance to clarify the sort of

framings or misconceptions that might end up being placed in other papers by philosophical philosophers.

And I think has become a little bit of a vicious cycle of distraction at times. And I think they're maybe trying to reclaim that ground.

So that's just my own strategic guess, but it looks likely that might be why this paper's been put together.

Well, great and helpful feedback. Thank you. Then the last parts are here. We attempt to redress this situation

by targeting each source of potential confusion. So they're going to identify three frequently asked questions, frequently raised concerns, underpinning frictions, and then redress them. And their aim is overall, we aim to increase philosophical understanding of active inference so that it may be more readily evaluated. And maybe we could

even add applied and developed. Okay, so would either of you like to just read the abstract?

I'll go for it, Dan. Okay.

Over the past 15 years, an ambitious explanatory framework has been proposed to unify the explanations across biology and cognitive science.

Active inference, whose most famous tenet is the free energy principle, has inspired excitement and confusion in equal measure.

Yes, it has. Here we lay the groundwork for proper critical analysis of active inference in three ways.

First, we give simplified versions of its core mathematical models. Second, we outline the historical development of active inference and its relationship to other theoretical approaches. And third, we describe three different kinds of claim, labeled mathematical, empirical, in general, routinely made by proponents of the framework and suggest dialectical links between them. Overall, we have to increase philosophical understanding of active inference so that it may be more readily evaluated.

All right. Cool. So nice and funny abstract. We all laughed. So the roadmap, how do they structure this guide? Is it meant to be read linearly or do you flip to a different page? I don't know. But we have added the page

numbers on this slide. It's on the bottom right, but it'll be on the bottom left. And that's what page we are in the guide. As if it were in paper in our hands. And it starts with an overview. And then there's a discussion on first the inference of perception or observations and hidden states. And then action comes into the loop. Section three gives a brief history of the free energy principle. And section four discusses the aforementioned dialectic of free energy and its rhetorical ecosystem, we could say. That's the part we're not going to focus on in this dot zero, but hopefully we're going to be able to go into each of these really important troikas in the dot one and two. Then there are some concluding remarks.

Okay. So first, how is the word model being used here? It's really good that the authors included this warning because we hear about modeling all the time, not just the photography kind,

the mathematical kind. So they write, let us begin with a warning. The word model takes on two distinct senses throughout our discussion. The sense more familiar to philosophers is what we call a scientific model, a representation of some possible or actual system, which a scientist uses to reason about or

discovers features of that system and related systems. By contrast, in the active inference literature, a narrower sense is typically meant. What we will call a generative model. That's what we're going to be specifically talking about. This is a mathematical object with applications in statistics and various sciences. Our simplified models of the free energy principle are scientific models. So models is a big category, but we're going to be talking about kind of a narrow sense of statistical models like linear regression models, but instead of a linear regression, it's a different kind of model, not at the exact same elevation of the linear model or taxonomy, but like in that category. And then note further that some scholars opt for a deflationary stance on generative models, using them only to describe the dynamics of agents. So descriptive modeling. It is an open question whether this kind of model building precludes any form of scientific realism about the relation between the model and the target system. These issues are discussed in section four and also have come up several times, including with most recently Majeed Benny, but model based science and what does it mean to use and make models and maps and territories and all of that. So any thoughts on models? Because that's going to be one of the key terms going forward. What would either of you say would be the best way in layman's terms to show the difference between what they mean when they talk about an empirical model and what they talk about in terms of a generative model? What sort of real physical material things could we use to make clear the difference? Because in the last stream we did a good job of taking columns and showing how they were discrete. How would you guys describe this? Yes, Stephen.

I think one of the ways this distinction comes about is the difference between the difference between most systems type approaches and non-linear dynamical systems. Generative models are tend to be coming from a sort of a swarming, it's like a swarm of bees creating and evolving as compared to something which has been given a perspective and drawn out somehow in our environment as in a scientific model. Now the fact that generative models are being applied in the kind of scientific way, inverted commas, or are they? Are they being used in a more instrumental stroke philosophical way and they can never be fully science is another question. But I think that the difference there between this kind of biological generative sort of process and something which can be given a perspective on and drawn out as a defined type of model is where there's a kind of a clear difference. I don't know how others see that, but I think that that also might be useful.

All right, Dean, here would be my answer would be two that are on the extreme end of the yellow type scientific model would be like the standard model in physics or a working model of renal function, like kidney function. So it's like very transdisciplinary and includes a huge number of domains and kind of logical motifs and has axioms and a specific history and software toolkits and all of that.

And then we're going to be talking about a kind of model that's basically a parametric statistical model. So like a Gaussian model takes two parameters, the mean and the variance, and that can be written analytically, like with equations, or it can be done computationally, but it's only ever going to take those two parameters and it only ever outputs a certain family of distributions. And so it's like parameters

in distributions out and the parameters can reflect like the connectedness of a graphical model too, but you get parameters in, which can include model structure and then some type of output, but it's it's closed-ended within that paradigm versus the model of renal function is like this open-ended question. And I think you hit on something really, really critical there. It's distributions out, right? So something falls out when we're talking about a generative model. Whereas when we're talking about something empirical, we tend to keep the focus stabilized, right? Like this, so the process is fundamentally different. And so again, when we're talking about in sort of layman's terms, what the difference would be, I think you hit the key word out of that, which is something that's in and stays in versus something that may start in one place and fall out in a completely different place. With is that an expectation we should have as a difference between these two?

Yeah. Just one thought then Steven would be both modes are so important. It's like yin and yang or left and right hand or convex and concave. And our discussions do move regime of attention from like the generative model to the bigger question. And in some ways it maps onto instrumentalism and realism too, but we really need to know when the pendulum is swung one way and when it's

swung the other way. And so we can keep an open mind and they're all happening all the time. That's both eyes open. Like you're always saying Dean, but we also can't go too hard on just one side. So Steven?

Yeah, I think I love what you said there, Daniel, about the axioms. I think in generally in science, you try and have reproducibility and you have tolerances and you say how much accuracy and what's this tolerance is on something, but there's a general recapitulation of similar types of results again and again and again. And the same sort of thing comes up. It has been there in psychiatry and psychology, you know, with, with, and that's where the, the idea of syndromes being classified as opposed to symptoms, which must be much more fluctuating, more variable. So you could run a generative model and it can come up with different results with the same input and that. Now there will be some kind of characteristic patterns that generative models do reveal, but that there is a point where it's quite different. And I suppose when, when that difference, I think there's almost paradigm based difference, but maybe there is some point where they blur. I don't know.

Yes, Dean, and then we'll go onto the paper.

Yeah. So at the end of the last live stream, one of the thing comments that one of our colleagues blew me was that we have to be able to hold two things up at once, which sang to me because I think that's what you're saying, both of you're saying right now. I think if we look at, if we, whether we're looking at a starling or a murmuration, whether we're looking at an ant or a colony, we can keep something in, inbound, or we can have something fall out from that because it's basic research. Like we weren't expecting something repeatable. That's, that's how I would distinguish it. But again, I just, let's get onto the paper, but I think that's important that we get that established first. If we're talking about guides.

That was the big epistemic disclaimer, and that's what they put in their paper. So we allocate our attention as well. Okay. So onto the details of the paper. So in 2.2, they introduce this

simple model of inference with  $W$  and  $X$ . So this is figure one, there's  $W$  for world. That's the hidden world state. And then there's  $X$ , the observed data, like  $X$  marks the data. And you can think about like, where do you estimate where the sun is in the sky? That's actually unobserved, but then there's going to be all kinds of observed data coming in, or we'll talk more about the coming in versus the generative model later, but the observed data are what are being modeled as being observed in that sort of sensory way. So figure one is the basic model of inference. The inference problem addressed by the active inference framework. What other kinds of inference problems exist? What problems do or don't have this form? What happens if there's no line with  $W$  and  $X$ ? So of course it can get more elaborate, but within this exact format, like  $W$  might output  $X$  and  $Y$ . And that'd be like getting two numbers back in the computer program instead of, or generating two numbers in the computer program rather than just returning one. Or you can imagine  $X$  is like a vector, but we're within that specific model framework. We're not in the realism of agent environment. We're just in the statistical framework of hidden latent variable, observable variable  $X$ . The unobservable state is assumed to cause observable data. And so this is a little bit of a connection with dynamic causal modeling, because statistically we call that causing just like the proportion of variance explained by a PCA. It's like given very active terms. It's, but it's actually not causal in the mechanistic sense. And that of course has been to no ends of confusion with how statistical conclusions are interpreted as mechanistic conclusions about the world. And so is it cause in the sense that we mean it in one sense, or is it associated in a causal model? And that's kind of the causal in dynamic causal modeling, that's functional connectivity. So that's the realm that we're talking about, not within the world environment interaction. Stephen?

Yeah, this is, this is actually interesting. They've gone back to a pre-Markov blanket kind of formulaism. And it raises that question then of what does it mean to infer? And it can be that we infer things because that's what we get data on. So it talks about, well, what is it that we can even use to understand? And what are the assumptions about what's actually important to read? And those two things, I think this leaves both of those things open. And that, that, that, that actually is a bit of a bridge between the scientific and the generative processes.

Cool. So if we were modeling like height is not measured, but we observed weight, and we're going to do some regression of like the unobserved and the observed, that is the type of model that we're going to be talking about loosely. And someone could say, oh, but this other factor influences too.

Yes. That's the realism claim. This actually matters in the real world and it very well may, but we're talking about measured models relationships, Stephen.

Yeah. I think this is also really helpful to be able to look at paradigms because there's a difference between say the paradigm of psychology from the perspective of looking at classification and problem identification and treatment and coaching psychology, which is about not so much trying to diagnose what's there, but what sort of actions someone's trying to take and how you might help them with those actions to reach outcomes, which may vary moment to moment. And I think, you know, in some ways that, that, that's, this is maybe the level you have to go back to because otherwise the noise just swamps the

discussion.

It goes deep on both ends, that action orientation, the pragmatic turn ecological psychology is going to connect deep.

And this is going to be deep, but in a sense, narrow, because we're just talking about statistics here.

So take statistics courses and learn it because this is kind of where it comes from here.

Okay. Onto the example of active cat friends. That's, that's not actually what they called it, but that's the cat example that we're going to be discussing.

We're going to imagine that you have a cat animal that spends its time in either the kitchen or the bedroom.

When it's in the kitchen, it often meows for food. When it's in the bedroom, it often purrs loudly.

Okay. Suppose you tally the proportions of the times your cat is in each place and making each noise.

The results might look something like, okay, but the cat also goes somewhere else. Okay.

Not in the model or this other factor or this other sound. Okay. Change the model, add another column.

It's totally fine. It's a fork on the GitHub or the discussion, but that's the difference between kind of modifying the model structurally and using it parametrically within the narrowest sense of

model and then keeping the discussion alive about that broader sense. Like what Steven is talking

about, like a model of psychology or a model of complex adaptive systems. So one thing to note is

that this idea of setting the prior in a Bayesian sense from observed data is called parametric empirical

Bayes. So parameter, parametric, we're setting parameters, like frequencies of things happening.

And it's empirical, which means observed. It doesn't mean like the only truth. It just means the measured

values. And then it's Bayesian as we'll talk about. So let's just say 10 or a hundred or 10,000 measurements were made and they come out to these numbers. 60% of the time they're in the kitchen and 40%, the cat is in the

bedroom. And so that's like summing across that's called a marginal probability because it was in

the margins of papers and they marginals have to sum to a hundred percent because something has to happen

in the model. Oh, but the cat can be in the roots. Okay. Yes. But within the model, it has to be this way.

Then that's what makes a probability. Then 50% of the time there's meowing and purring, but as the numbers show the location and the sounds, they have a statistical association. There's some divergence

that could be tested for statistical significance, depending on the number of observations. But that

is what the numbers in this table are reflecting. And both columns and rows sum up to a probability. So

they're all like proper probabilities. Then how do you infer about this? Like if you were to

observe the room, how could you reduce your uncertainty about what sounds you'd hear?

And then if you'd hear sounds, how could you reduce your uncertainty about the room?

Many philosophers will be familiar with one famous method for solving this problem, Bayesian

conditionalization. This method can be stated as a principle saying how an agent using a model,

$P$  of  $W$  and  $X$ ,  $W$  world state hidden  $X$  observed data ought to choose their beliefs  $Q$  of  $W$ ,

$Q$ , that's the one we control upon observing data  $X$ , what to believe now conditioned on incoming data.

Okay. Any points on that?



Just, they did a really good job of explaining it.

Yeah. I'm waiting for the next slide.

Yeah. Okay. Please continue here.

So from the, from the actual paper, the table describes a joint probability distribution, which we've just explained, where  $W$  ranges over possible cat locations, either the kitchen bedroom and  $X$  ranges over the possible cat sounds. Again, very clear, straightforward. From the next side, usually the science, scientist, not the person who is trying to build a generative model aims to improve the accuracy of a generative model of some real world, world phenomenon, which would mean improving the accuracy of  $PWX$ . This learning task is relatively difficult. So I wanted to kind of parse what relatively difficult meant. And I'm not going to read all of that stuff there in red, but essentially what I want to kind of state is they, the authors here said, this is simple. Don't, don't go to that place of high complexity quite yet. Let's just see this for what it is. So that what Daniel said 90 seconds ago, which was go take a bunch of statistics courses. Doesn't have a bunch of people running off and screaming into the night.

There is, there is a complexity to this, but it's not the kind that will swamp you. What they're basically trying to say is let's just slowly work our way into this without every hit, all your historical grievances around statistics trapping you before you even really set out set sail.

I'll add one note on that. This is like learning how the horse moves in chess or learning how the castle moves in chess. And so it's possible to get super connected to your internal life narrative and the game of chess and people who would have no problem losing in connect four or checkers will feel very engaged emotionally affectively by chess. It just something I've observed empirically. Somebody else might have different priors on that. And so this is a way of just starting with how the pieces move. And then as we're finding out,

we're going to figure out how the pieces move and then we're going to connect them and do all kinds of fun stuff.

Steven?

Steven Yeah. I think like you say, having the ability to stay with the pieces, do lots of fun stuff and look at the the way things are generating with that data is different, for instance, to saying, oh, the meowing cat is an assertive cat and the purring cat is a docile cat, which could be this kind of imposed higher order model. And then everything's trying to fit to is it a docile cat or is it an assertive cat? But it's flattened out all of this, you know, it loses this ability to generate.

Steven Professor Helen Longino has a book called Studying Human Behavior and Aggression in 2010.

That's like, once you define, oh, there's a significant difference between these two groups, or in the count of this

behavior per minute, and they've been pre labeled, it passes the modeling in the broad sense into modeling into the narrow sense.

Steven Professor Helen Longino has a big difference between the groups. So this is reifying our understanding of those categories. So it's really important. So this is going to be where the base enters the picture.

Steven Professor Helen Longino The posterior, which means afterwards, is calculated about how should the distribution  $P$  be updated as new information comes in. So just like the last sentence in red, what's the difference between the groups?

Steven Professor Helen Longino The posterior, which means that the last sentence in red, what's a belief now conditioned on incoming data. So now that is going to be symbolically or graphically described. So probability, the  $P$  distribution of world states conditioned on vertical line, new data coming in. And there's Bayes theorem.

Steven Professor Helen Longino Theorem. Other videos and other channels and groups will cover Bayes better and more comprehensively. But here we can suffice to say that first we can write it in words. Like what's the distribution  $Q$  that we control? The probability distribution of us thinking that it's in the kitchen.

Steven Professor Helen Longino Theorem That is the prior probability of it being in the kitchen conditioned on it or it's conditioned on it meowing. So how likely is it to be given that it meowed? And then the numbers that we just looked at can be plugged in to that verbal equation and a number is going to come out.

Steven Professor Helen Longino And so we heard meowing. And if we just looked at meowing like four times in the kitchen, one time in the bedroom. So four to five times it was in the kitchen, not the bedroom. And then here's an equation that does literally that. It just looks at the column and it does the conditional likelihood or the posterior of the world state after the data come in.

Steven Professor Helen Longino And then there's, I think, did anyone want to say anything here? There's this, it's

Steven Professor Helen Longino Learning task is difficult. I think we already kind of mentioned that.

Steven Professor Helen Longino Yeah, and the whole thing is, is that most people, I think, I'm not going to say philosophers, most people don't automatically go to being able to figure out what the probability is. But I think what these authors are pointing out is,

there's a way of being able to show that there is a probability in play, if you are not certain. And this is how you might split those probabilities out. That's all they're trying to do.

Steven Professor Helen Longino Okay, so now we're going to go from, as we've seen in other papers, the sort of exact Bayesian approach. Most similar paper to this, in this sense, is probably Axel Constance, number 34, with a bacterium.

Steven Professor Helen Longino We first looked at the Bayesian bacterium, and then we looked at the variational Bayesian bacterium. And then in both cases, we looked at how basically, sometimes new data could come in, you could still make wrong decisions.

Steven Professor Helen Longino Whether you're doing exact Bayes or variational inference, you can still make bad decisions as data come in. So this isn't a Panglossian paradigm. This is like a statistical tool that's already been detached from the philosophy, or at least sort of docked at shore. And then now also we're breaking the assumption that it must go the right way, or that it will model real world systems, especially.

Steven Professor Helen Longino So again, back within the narrow sense. Sometimes the specific distribution of the world states conditioned on data coming in is intractable, so that distribution cannot be calculated.

Steven Professor Helen Longino This usually happens when the state space is continuous rather than discrete, which came up on model stream number 5.1. If there's only two decisions left and right, all you have to do is do two calculations and then compare which one is preferable.

Steven Professor Helen Longino Whereas if there's a steering wheel or a trim tab, then there's an open ended number because 87.1 degrees and 87.11 degrees might be very different in their long term

consequences. So continuous state spaces are very challenging from a sensory as well as from an action perspective.

Steven Professor Helen Longino In these cases, what is needed is a way to choose  $q$  of  $w$ , the distribution we control on the world states to make it close to  $p$  of  $w$  conditioned on  $x$ .

So  $p$  of world states conditioned on data is the hard one to compute exactly, and this is going to be choosing a distribution that gets close to that other one.

Steven Professor Helen Longino When the problem is formulated by statisticians, we usually begin with a family of possible distributions  $q$  and search for the member of that family which lies as close to  $p$   $w$  conditioned on  $x$  as possible.

So just like a linear model, it's like conditioned on this being a linear model, we're going to find the best linear model or conditioned on the quadratic form, we're going to find the best quadratic form.

Steven Professor Helen Longino This is like a little bit more general conditioned on the family that  $q$  is, we're going to do basically linear regression fitting, but not exactly linear regression.

How do you make those distributions converge or align as best as possible, we can do this.

Also, given that a structure has been chosen.

This is the fitting part.

This is not the structure learning unless the parameters about structure.

We can do this indirectly by using a measure of inaccuracy.

Active inference employs a measure of inaccuracy called variational free energy labeled  $f$ .

Because it is a measure of inaccuracy, smaller values are better than larger values.

So it's like Frisbee golf.

And the lower the value is, the lower the difference is between the actual  $p$  that you would have calculated perfectly and  $q$ , the way simpler, lower dimensional model that you can control and can save in RAM.

So that's described here.

Variational free energy captures two sources of inaccuracy that we're going to go into in the next slides.

In belief and dictates how they ought to be traded off against one another.

The two sources which we're going to explore are overfitting and failing to explain the data.

So those will be introduced and discussed soon.

Steven?

Steven?



yeah i think this also if we just to connect it to that scientific standard models and this type of work is this is looking at contextuality it's like it's like what's based on where the starting point is that the probabilities are being picked up on you start to roll out the um statistical results i what sort of meowing is actually happening and as opposed to a model where you're modeling a car engine and you want to know what energy it's going to give out a particular moment it's going to do that whether i'm watching it or not it's just the same basic model applies independent of the observer so i think that's uh also useful here yes like getting the model to this stage like depersonalizes it in a sense because that model can be just transferred and used in a like another instrument okay so here's those two sources of inaccuracy that they described the first that we're going to discuss is overfitting the cost of overfitting they write can therefore be measured by checking how far  $q$  the one we control on world states diverges from  $p$  the first term of  $f$  free energy is a measure of that kind this expression two is also called the relative entropy or holbach liber or the kl divergence so this is like the first half of the two-part equation that will constitute  $f$  and it looks like this graphically symbolically but here's um a kind of cool way to think about it if you were trying to fit a single hump into this two hump model in the empirical distribution um so you're fitting a family that is going to get sort of coerced into one of two extremes either it's going to end up fitting one very closely like the higher one if it's like mode seeking and it will have no probability density the blue line onto the other hump or it'll conflate them in a sense into two um uh kind of subpopulations blurred together and have a solution basically that kind of goes between them um kullback divergence is a way to fit distributions optimally given this kind of challenge which exists for the one hump fitting two hump it exists for the two hump fitting three it exists for the one fitting 50 it's a general statistical problem and kl is a method that helps bring that blue line as well as it can on a trade-off frontier into alignment with the black one okay stephen yeah i suppose this also gets into the reason when this is useful is when you're in complex uh non-equilibrium situations where things are fluid then you're going to hit these situations and overfitting is quite common social science is being one of the areas where we see these problems where it's equilibrium based and it's fairly clear then you don't really have it's not really as applicable so it sort of speaks to where this type of approach is applicable i.e where there's this kind of fluid ambiguity in in in the situation okay thank you how about the second source of inaccuracy that is failing to explain the data mathematically they write explaining the data means assigning high probability to events  $w$  that make the probability of  $x$  high the penalty for failing to explain data is captured by the second term of  $f$  which looks like this um higher terms of the actual value are matched with high value um with high

values on the distribution we control  $q$  to keep this term low so this is kind of like if it were a linear regression and all the data points were lined up and the line just went right through them then it would be doing really well on explaining the data and that would be given a low score here whereas if that weren't the case it'd be flipped dean so when i was doing my work i didn't have access or maybe i wasn't looking hard enough for the kl divergence so i can remember even on here and some of these live streams saying i have no idea what that even means i have a better idea now but it's one of the metaphors that i tried to use was in terms of avoiding overfitting or failing to be able to explain the data was so all those horizontal and vertical lines creating a mesh acting as a filter how far apart do we set those what gauge do we set our mesh for in terms of what we want to have stay above the mesh and what's passing passing passing through now again maybe that's not sophisticated enough for a lot of philosophers but to get the basic point down that's the kind of example i had to provide to be able to give people some sense not just that these lines are now rigid and the 20 is on this percent and the 40 is on this percent and never tell shall the two pass again but the lines actually move

thanks a lot so you're so right about that the example with the active cat friends and the variational cat friends is like discrete by discrete is four quadrants so in some ways that's like the simplest model right it could be two continuous variables it could be like the volume and then the cat's position on the x-axis in a room now imagine if it was the x and the y is continuous or something like that but just even two continuous variables and you're totally pointing out where it's like you're gonna put the points around whether you put them right in the middle of the four quadrants or whether it's more scattershot or something in between and then there's two parameters there's like a linear regression through the points and then there's how fine the mesh is and you're observing like pixel densities and doing a regression through the pixel densities and so you could have a super continuous situation it is a continuous variable inside the bedroom but then in the model we just looked at it's in one quadrant and we talked about that when we talked about um serval's paper and how it was just the park and the cafe and yes there's locations within the park but they weren't within our model so we didn't deny the reality of that physically we just were modeling statistically something specific okay steven i mean this the filtering could also be relating to the sort of temporal sampling rate because if you're measuring every millisecond if it's meowing or not it becomes pointless there's a kind of a making sense rate at which the um the entropy is being converged at and and as we know in the brain we seem to have multiple

levels of that okay awesome next few ones we're going to go through just kind of quickly and specifically so let's put those two terms together this is the overfitting and the over and the failing to explain the data variational free energy  $f$  the non-metaphysical version is the sum of the penalties for overfitting and failing to explain the data it's those two terms that we just described added together so that the best explanation in a sense is the lowest on both of them it's not overfit it fits it well and it doesn't do it any more than that so you know simple as possible but no simpler any other number of quotes that's kind of this equations it's not the only way it could be written it's not the only thing that fulfills that process but it's something that can be used and it's very tractable

as it turns out um this free energy has tractable computation 13. page 14. so now it's possible to actually use that tractable approximation and do decision making and here is where they have the decision making um here argman means choose the distribution  $q$  that makes the following term as small as possible so this is where  $f$  goes from kind of being like a um a set of pipes with nothing really running through it to a specific finite set of tested alternatives those are affordances in the case of action selection they're discrete in the case of a discrete inference like is it in the bedroom or is it in the kitchen it's continuous in the case of a continuous inference problem like how bright is it outside on a continuous scale but even for continuous things sometimes we just do one through ten or one through a hundred or one through a thousand and twenty four so discrete state spaces are really important even if there is a continuity to the world so those are the kinds of computations that variational inference helps perform it takes something that's descriptive and moves it into a decision making imperative just like the  $L_2$  norm or least squares is like a decision making imperative for linear regression this is like an imperative for model fitting in this framework um the free energy principle the form here inference is the same as that of bayesian principle discussed earlier in both cases you perform a calculation and set  $q$  of  $w$  equal to the resulting value the difference is that bayesian principle counsels a direct calculation via bayes theorem in contrast free energy principle counsels what might be called an indirect calculation happily in practice this can be done by trial and improvement rather than trial and error so that's axel constants bacteria the bayesian and the variational bacterium combined with going to the bottom of the bowl on the smooth gradient descent landscape with a straight arrow utility and the solenoidal epistemic component various algorithms for finding  $q$  the distribution we control are available depending on the details of the generative model classic citation that gets referred to a lot

one of the developments that prefigured active inference was the implementation of such an algorithm in a neural network first in 2005. old citation classic we're seeing that again in model stream 5 in alex chance's work and of many others that this format even long before active inference brought it onto the scene in

our little side stage using neural networks to fit variational bayesian methods was a common technique in machine learning stephen so when we say trial and improvement rather than trial and error it's basically saying

that the error can be used and utilized in an ongoing way it's not like you have to start from scratch and come up with another attempt from scratch you're you know you're using what the same basic architecture and just keep running

it's so true i never really thought how demoralizing trial and error was like how should we get stronger oh well you're

going to try and then you're going to fail when error is perceived as a failure naively but trial improvement would be like we're going to

try and we're going to improve so yes of course error is implicit in that dean yeah and i'm glad you used the bowl analogy because one of the things that i read and i was actually the one to highlight a trial improvement thing because for example if you had the bowl filled with water and a ping pong ball

floating on top and you had to drill a hole because you wanted the ping pong ball to settle at the bottom of the bowl at a certain time you could continuously improve by adding more holes of a certain diameter until you were able to get that flow rate and have that ping pong ball arrive at the bottom of that bowl at the particular time at the discrete time that you were looking for that's not an error thing that's just uh we'll continue to get a little bit closer and closer we'll figure out what the dosage is and i agree with you a lot of people think it that the basic research part of it is oh i failed no a lot of this stuff is i test it and then i test it again and then i oh i came really really close so now the difference is really low which is what these authors are talking all right i'm just going to continue on because we do have a ton more to do so that's still on the perception side we're not talking about where we're going to go walking we're talking about the sound of the cat and where the cat is located in the inference and so we're still within this empirical variational catference area this is the bowl this is fitting that distribution the conjoint distribution that's the free energy distribution and that is the one that's bull like so here is the variational free energy the black line and it's basically the composition of two factors penalty for overfitting and the penalty for failing to explain the data and each of those have a certain distribution underneath them in this setting so their combination is like  $f$  plus  $g$  of  $x$  it's like  $h$  of  $x$  it's just adding functions together so it's just another cost fitting function for the really specific kinds of models that we're talking about here let's bring in action so now it's still the same case of  $w$  and  $x$  going to the agent now notice that the agent wasn't drawn in the previous model it was just  $w$  to  $x$  so that was um probably a graphical as well as somewhat of a conceptual uh simplification like i mean these measurements it shouldn't matter who's

observing them right oh wait quantum it does but now the agent can also take action  $z$  so  $z$  is going to be the whole question of control theory and cybernetics and action the previous section dealt with inference rule how to choose  $q$  of world states beliefs on world states this section is about acting now suppose you can perform an action  $z$  that will place the cat in one of two rooms by changing the hidden state  $w$  you can indirectly change future values of  $x$  or at least change their proportions or their likelihood so decision rules stem from measures of preference because if you don't know where you're going you're lost or if you don't care about the two things then it doesn't really matter one of the confusing aspects of active inference is that it treats the statistical model  $p$  the one that is the actual distribution we're trying to get close to and this is like the key point that we'll be returning to for our whole life that model  $p$  is a measure of both probabilities and preferences at the same time and that's going to be what we continue to talk about because it's one of the most important points that this paper makes clear in a way that other papers um haven't really perhaps harped on in exactly the same way let's look at figure three again look at the yellow part active inference employs a controversial dual interpretation of  $p$  of  $w$  probability of world states and probability of observations as probability distributions and preference distributions over hidden states and sensory states specifically deal with the red text again i don't want to take up a bunch of time but this we're having one of the authors or a few of the authors on we can probably pull this out a little



bit but for now that when we when we introduce preferences and probabilities it isn't just a second it isn't just a second consideration it can almost run away from us really really quickly if we're not really careful in terms of pulling back on the reins a little bit and really thinking about what does that imply so i'll just leave that for there for now great point thanks for sharing it succinctly

they write recall that free energy principle inference counsels choosing beliefs by minimizing a function that measures the cost of inaccuracy and that's because the free energy calculation includes both of those very uh features the overfitting penalty and the failure to explain penalty action selection is governed in the same way as inference remember we were talking only about inference when we were talking

about the variational methods now there's going to be a twist a slightly different cost function called the expected free energy so that's why we were previously just talking about the  $f$  variational free energy now we're going to be talking about  $g$  expected free energy labeled  $g$  the definition of  $g$  is closely related to that of  $f$  the interpretation of the two penalty terms changes as the formalism is updated to reflect the fact we are now making measurements over expected future states so this is not just measurement error there's the fundamental unknowingness of the future since future states have yet to be observed the agent must average over them to obtain expected values the penalties are associated with failing to satisfy preferences and failing to minimize future surprise overfitting preferences failure to explain surprise expectation preferences preferences when now or then let's look at how those play out in this sort of action selection through variational free energy through time aka the expected free energy formulation and how those two pieces satisfying preference failure and failure to minimize future surprise look now but this is sort of the elaboration or the cousin of this one this is like snapshot inference and now we're going to be looking at expectations through time

okay so first feeling to satisfy preferences so again keep that simpler version in mind this is going to be a slightly different one in six

they even write compare equation two

um this is make i hope failure feeling to satisfy preferences given a clear enough term but it totally brings in something different which is

something about the agent as dean wrote so there's big implications of course because there can't be a non-preference driven action selection

in any useful sense whereas the variational one was just like given the scatter i want the best mesh and regression

that one it does embody implicit assumptions about the world and so on but this takes it to a whole another level

and here's where that controversial dual interpretation comes into play not only is it unusual to treat  $p$  as a preference distribution

it is unusual to treat the goal of decision making to produce a distribution that matches that distribution rather than maximizing expected utility

so you don't need a preference distribution other than more for reward learning whereas here

we're trying to realize our preferences and match distributions rather than maximize for minimizing the

divergence from

these realistic but optimistic expectations so perhaps it is best to keep in mind that preference in this sense might mean something different than utility in the traditional sense

the difference

steven

yeah without going back into that but the whole point of having this matching distributions rather than expected utility which effectively again becomes kind of like equilibrium states places where things settle and can

be measured and they're kind of stable you're in this kind of realm of a more fluid flux type process

um so i think that's that's one thing and i think the other one is um this focus on the future the focus on

predicting and things going into the future and how to make sense of that rather than looking back

and trying to which is often what science is doing is and explain retrospectively

in the same way psychology is often trying to explain what it sees before it and coaching psychology is trying to

see what can be done to get closer to something that's more suited great point thank you it really embodies

the forward-lookingness rather than the optimal reward or prediction on previous observations all right so failure to

minimize future surprise this is the other term failure to satisfy preferences now failure to minimize future surprise

formalism seven one of the tenets of active inference is that agents should act this is in the normative stance to ensure that future data are not too surprising and so here is the formalism as written

this is the failure to minimize surprise in addition to conditionalizing on  $z$   $z$  is action so we know that

the vertical line is conditioning on and then  $z$  are the affordances like the action states in this simple

example we're not going to go into the markov to the policy into the action state yet this is from the

simpler cybernetic or sort of agent environment framing that doesn't distinguish that as clearly

but  $z$  is just an action

that the agent engages in

the failure to minimize future surprise is conditioned on  $z$

so it's not that there's some sort of world that we're not altering and then we're doing some strategy in that in alterable world

leading to this total

ad hoc way of integrating the outcomes of action into the niche

the world states in the future have expectations that must be calculated as expecting on policy selection

so it's not just that some policies have effect and some don't it's that the inference of

the future failure to minimize surprise is conditioned statistically in the algorithm on the choices

the future of the future

so that is one of the cool parts about this calculation

um and then

this expected free energy okay steven yes go ahead

that also brings in an element of this contextuality it says it's what sort of choices

were made then in context  $x$  or well let's not use that context right it's important and uh

it changes things and you can't do that if you're taking averages um and sort of pre-built models

okay so now just like we kind of looked at the two parts separately and then summated into

$f$  the variational free energy and then looked at those two parts here we've gotten to equation

eight in the paper which is  $g$  which is a function of those two distributions  $p$  and  $q$  as well as action

and here are those two pieces that we just discussed and then they write the third input to  $g$

$g$  is  $z$  rather than  $x$  so not observations but action as mentioned above this is because we are

calculating the expected value over possible future sensory states rather than inferring on

the basis of a sensory state that has just occurred okay so that's exactly what stephen said  $f$  is really

good for sensory inference it's about given the just observed sensory data what the estimate of the world

should snap to but action is totally different not just because it entails preferences like dean raised

but also because the consequences of action and the future are unknown so the distribution for

planning as inference for action as inference rather than for perception as inference needs to require

conditionalizing on action not on sense even though sense does come into play it's a little bit like

hidden away here's that what we're getting at which is the sensory distribution noticeably not here

but on the right side only and then the function that we're minimizing on is action inference active

inference as with  $f$  the measure  $g$  suggests a principle free energy principle in terms of action not as

inference is minimizing the free energy on action selection that can be read as an approximation of not

optimal bayesian sensing but optimal bayesian design of experiments and optimal bayesian decision theory

dean

and i think right here is i'm just pausing for one second i think that we're going to get into some more

evidence of this but i think this is the first moment when we can say active inference and free energy well not free energy

and this is the first moment of the time that we can say active inference does not necessarily constrain itself to being just a framework

it's actually a filter as well because of the active part it's not stable it's constantly being updated because of that active piece and the preference piece so we could call it a framework but i don't think we're doing it justice there's a framework aspect to it and there's a filtering piece to it and so that's why i talk about search fields all the time

so i just want to drop that seed now so that when the authors come on we can talk about that a little bit more in terms of the context that stephen mentioned about guiding

awesome thank you so now we're going to return to the active cat fronts so let us present a solution to the cat example for the problem to have a determinate solution we need the conditional distribution world states condition on action  $z$

that means that the consequences of action have to be estimated if we put the cat in the kitchen it usually stays there so this is again the two pieces that action introduces into the puzzle is the question of preference otherwise why bother and the question of the consequences of action so the red is a statement that's

empirically observed from observed data about what happens what happens when you are estimating location and the consequences of action is the question of the consequences of action and the consequences of action.

so the red is a statement that's empirically observed from observed data about what happens what happens when you are estimating location conditioned on action this is that  $q$  of  $w$  conditioned on  $z$  and so you can see when it goes into the kitchen it stays in the kitchen nine out of ten times whereas if it puts in the bedroom after some period of time like steven brought up not one millisecond later but one hour later one minute later it's model specific so that's a

it's time scale friendly it's time scale friendly it's not time scale free and then you can compute numbers having to do with action selection within this model it is worth restating how unusual it is to interpret  $p$  as a measure of both probabilities and preferences but i mean it's the letter  $p$  there's nothing wrong with treating a distribution as a measure of preferences distributions don't demand to be interpreted as probabilities after all but what is unorthodox in what church and in need of justification is giving the very same mathematical term two different interpretations within the same equation so that's the two eyes at once kind of looking back and then they go into a little bit more detail about what that actually means we are not aware of proponents of active inference taking this interpretive line but it appears to be a viable option so just awesome

and clear writing and clear writing and drawing something out through the re-understanding and the communication which happens synchronously and asynchronously

one of the ways proponents of the framework turn to this unusual interpretation to their advantage is by casting action as a form of inference so here's from a buckley citation and that's why it's called active inference

and we just kind of a mechanism underlying minimizing expected free energy is formally symmetric to perceptual inference

formally symmetric overfitting failing to satisfy preferences failure to explain the data failure to minimize expected surprise on future data that's why it's the only one that has the  $x$  in there

rather than inferring the cause of sensory data an organism must infer actions that best make sensory data accord with an internal representation of the environment

statistical representation not the debate on whether organisms have representations

2.4 simple model of selection this is where the markov blanket is going to enter in our model  $x$  and  $z$  are the inputs and outputs of the agents  $x$  observations  $z$  actions

the set  $x$  and  $z$  is the agents markov blanket the term is derived from judea pearl's work on inference using bayesian networks

pearl 1988 classic citation other live streams we talk more about markov blankets so we're not going to go into it too much more here

but it's definitely the tip of an iceberg if you want to check out more but here in the sense required here

a markov blanket can be understood as the set of nodes that screen off the agent from nodes considered external to it

screen off the agent or screen off the internal states of the statistical model

another toy model will help illustrate consider an agent whose service temperature  $x$  can safely lie between negative 4 and 4 units

temperature is continuous okay but it's a discrete model

and so here death is coming into the picture on either end too hot and too cold  
so this is going to be like variational catference but now it's a freezing and a burning end of the room  
and the cat or the robot cat is going to be making decisions about what to do  
so the external world state is  $w$  the temperature and then the perception is going to be  $x$  the observation  
notice that the value of  $w$  does not affect the agent's preferences  
so the preferences for living or for homeostasis or whatever regardless of the observations in that snapshot moment  
but not precluded over longer time scales  
all the agent directly cares about is its surface temperature denoted by  $x$   
that's why the two rows are identical  
so these are sort of the observations and the different actions that can be taken to move  
between one end and the other perceiving only the temperature  
the agent tries to guess how to act how to stay within its expected range  
and then this is sort of the intuitive pseudocode  
when the temperature is high you want to go towards the lower end  
when the temperature is low you want to act to go towards the higher end  
and so here is like the simple example this is kind of like the bacteria that has the right priors  
versus the one that's just deciding randomly or without any resemblance of relationship to the other observations  
and so here is the smart agent that's just keeping its free energy every time it gets up as it moves away  
it acts to bring it back down into alignment  
and then here's um the agent's control over its external state is 95 accurate  
so five percent of the time it slips and then that's what they say it's grasp slips  
the optimal control the handle has been lost because of a motor ineffectuality  
but that can be recovered whereas the randomly acting or just sort of like the sort of brownian  
walk diffusion process ends up dying  
okay dean  
all i was going to say here is is did it create did a great explanation  
and if this was your first encounter with the  $w$  and the  $z$   
and the  $x$  you'd have to really really slow this down  
because the first time you encounter it it it takes more than one pass  
you might have to go over it a third and a fifth and a seventh time for it to actually make sense  
because there is we all have a certain capacity in terms of the amount of variables we can juggle at once  
and i was when i first looked at this it it made sense to me but i could sense  
from somebody who's maybe coming from a philosophical background  
and not necessarily having the same degree of statistical  
um hands-on experience as maybe the three of us do  
this would be a moment where you'd really want somebody to hold your hand

that's what community and act in flab is for  
because then it can be like an interaction and every question is welcome  
so as long as there's attention in the game  
everybody's going to make it steven  
so this also does give an indication that  
with variational free energy having some sort of perturbations can be useful  
particularly if you have multiple um parallel kind of sources of information  
being integrated  
so having some noise is a good thing right it's not it's actually gives you a way to start to  
make inferences one question is there  
that on the left there's no real learning happening  
no it's just but it still makes it even without the learning component it still is more stable  
um yes it's not doing parameter updating  
it's just like um a previously learned association  
between the temperature and the direction to move  
and also notice like the observation of sensory is assumed to be perfect here  
so there's many layers which can be brought in  
but this is kind of like the kernel motif  
okay  
the correspondence between high values of  $f$   
and life-threatening states lead to a third form of the free energy principle  
so we had inference and action  
perceptual inference  
then we had action selection  
and here is free energy principle selection  
any system that survives long enough  
will act so as to appear to be minimizing  $f$   
that's the first time we got any discussion of  
far from equilibrium  
thermodynamics  
of anti-dissipative systems  
resilient systems  
anticipatory systems  
except indirectly through action  
but this is the sort of selective  
darwinian side of fep  
okay  
of course there's a ton that could be said here

this is not a normative principle  
not a suggestion to agents  
regarding how they should perform inference  
but a means of describing how agents behave  
axel's paper  
minimizing free energy is not living  
alone  
but living systems will be those that appear as if they've minimized free energy  
and then here  
i think someone wrote  
i just wrote that in there because  
yeah  
that's going to tell you what your autobiography is going to sound like  
given that you're still alive and can write one  
yep  
my last words are  
ah  
in recent work  
friston gives a deflationary interpretation  
on which agents  
do not in fact minimize anything  
but perform acts  
which can be interpreted as minimizing  $f$   
so  
see  
live stream number 34  
0 1 2 3  
that is the reason  
for the emphasized phrase  
so as to appear to be minimizing  $f$   
that's  
deflationary instrumentalism  
highlight  
despite this deflationary approach  
there is a link between this and the earlier principle  
agents subject to the free energy principle of inference  
perceptual inference  
ought to minimize  $f$

so if this is ought  
is tied to their survival  
then the normative principle  
has the same underlying justification  
as the descriptive principle  
so this is sort of  
i don't know if it's been  
noted before  
but basically it's equivalent to saying  
the evolutionary ought  
is and is  
so it's kind of like  
not from  
the fitness  
side  
the  
malthusian  
darwinian  
side  
but from the  
anti-dissipative side  
this is how things  
have looked  
it's like saying  
fitness isn't necessarily  
um  
being projected  
into the future  
but if you do  
do the computations  
fitness can be assigned  
to like  
different single nucleotide polymorphisms  
that have had different success  
and you will only see successful ones  
but it's a bit more complex than that in the moment  
so  
here are the three



piece  
the tale of three piece  
we had the tale of two densities  
what other  
what other jokes have we had  
in each of the three examples  
discussed in a section  
there's been a distinct role  
for the distribution  $p$   
and a distinct interpretation  
of each model  
narrow sense  
in our first model  
 $p$  was a generative model  
employed by an agent  
it was therefore interpreted  
as representing  
probabilities  
like where the cat is  
but now  
in order to introduce action  
in a meaningful way  
we had to have preference  
like I want the cat  
to be in the kitchen  
so when we brought action  
into the game  
we had to  
introduce preference  
what makes active inference  
similar  
but also different  
from other frameworks  
probabilities and preferences  
are represented by  $p$   
in addition to representing probabilities  
in the second model  
 $p$  measured the desirability

of certain future states

over others

I expect you to go to school every day

that's something having to do with action

if it's serious

it was therefore interpreted

as representing preferences

in our third model

p tallied the historical frequencies

of a set of hypothetical ancestors

fitness

it was therefore interpreted

as representing the fitness

of different states

so

supporters of the framework

often point to the third role

fitness

p h

to explain how p

can simultaneously

fulfill the first two

it's like

if that thermal bacteria

is

rocking its niche

it will have high fitness

okay

so

that is

where the three p's

get us

here's just a brief

look forward

so for any of these next slides

we're not going to go into

the actual

very nicely fleshed out

arguments themselves

but

Stephen or Dean

just raise your hand

if either of you have anything

to add on like

each of these

section headers

so

um

section 2.5

is extensions to the model

more things to learn

more ways to act

we've seen

adjectives added to

active inference

deep

sophisticated

contrastive

affective

what other adjectives

have we seen

that's sort of this

section of the guidebook

it's like

there's your um

usb port

what can you

have

added into it

or how can it be developed

and so

that's where we see

sort of the adjectival

family

of something

active inference

maybe even  
n  
active inference  
coming more to the  
philosophy side  
but also  
there's all the  
elaborations that we've  
seen of the actual  
parametric model  
in the narrow sense  
sometimes  
it's in both  
camps  
like affective  
inference  
has to do  
with  
a model  
derivation  
or metacognition  
as well as  
something about  
like the model  
in a bigger sense  
but  
we  
can recognize  
both those  
lanes at once  
Stephen  
yeah I mean  
one other one's  
inactive  
inference  
uh  
reciprocal  
active inference

there's a couple more  
like that's definitely  
uh definitely  
true  
just one thing I was  
going to throw in  
I noticed they talked  
about this  
and it's I know it's in  
the quote but  
they talk about any  
system  
I always wonder whether  
that's a bit of a  
that a little bit of a  
piece of a paving stone  
waiting to be tripped  
over  
um  
because I kind of feel  
like  
you know  
if you're going to  
say it maybe  
any non-linear  
dynamical system  
or nested  
systems  
of systems  
or things  
I just always  
think that  
that can be  
a little bit  
of a  
hazard  
yeah it inherits  
a lot of the

legacy  
ambiguity  
around system  
like you've brought  
up a bunch of  
times  
so it's really  
helpful  
so  
in section  
three  
notice how  
the history  
brief history  
of the free  
energy principle  
comes after  
what we just  
spent the last  
hours and pages  
on  
so it isn't  
a history of  
science perspective  
but it is being  
recognized as  
important  
and so the free  
energy principle  
is a modern  
incarnation of  
ideas that have  
been raised  
sporadically  
over at least  
the five  
decades  
it combines

traditions from  
physics  
biology  
neuroscience  
and machine  
learning  
and other  
areas  
especially  
the modifications  
and increasingly  
the applications  
and it's a  
bi-directional  
freeway  
it's not just  
two lanes  
in both  
direction  
it's going  
both ways  
so this is  
sort of the  
history  
which is fun  
because it's  
a recent  
history  
in some  
parts  
okay  
section four  
moves from  
the historical  
to the philosophical  
so section four  
dialectic  
the free

energy principle  
and related  
claims  
mathematical  
empirical  
and general  
claims  
section 4.1  
this is where  
all that  
mathematics  
groundwork  
pays off  
because in  
the author's  
words we think  
a great deal  
of confusion  
can be overcome  
by considering  
three kinds  
of claims  
first there are  
mathematical  
claims  
those were the  
ones that were  
just brought up  
in earlier  
sections  
so that's  
the theorems  
the scientific  
models and the  
statistical  
techniques  
in the more  
narrow sense



and by doing  
good scholarship  
we show that  
the core features  
absolutely  
predate  
active  
inference  
and there's  
less  
controversial  
than some  
might suspect  
however  
Fristin colleagues  
have since  
introduced many  
novel mathematical  
elements  
like the  
perception  
and action  
interpretation  
on the  
Markov blanket  
importantly  
claims in this  
category do not  
need to be  
interpreted as  
statements about  
real systems  
in order to be  
evaluated  
so part one  
is  
it's  
instrumentalism

part two  
is  
separate the  
parts that  
predate active  
inference from  
some of these  
more recent  
derivations  
even within  
the only  
technical  
area  
second area  
of confusion  
or the second  
way that confusion  
can be overcome  
by considering  
another kind of  
claim is  
partitioning off  
you know we can  
add like a line  
here from that  
mathematical  
everything that  
was in the  
section that we  
just discussed  
plus stuff that  
was more recent  
like 32  
there are  
empirical claims  
about cognitive  
and biological  
systems how

brains and  
bodies actually  
work  
these are the  
remit of  
cognitive  
neuroscience and  
biology so  
those are  
empirical  
biological  
claims  
third are  
general claims  
that typically  
abstract across  
a wide range  
or by class  
of empirical  
claims so  
that's like  
the sort of  
everything  
nested systems  
collective  
behavior angle  
so separating  
those and maybe  
there's more  
is really helpful  
because sometimes  
people say well  
this general  
claim is true  
because look at  
how this  
in the skin  
works well

what about the  
skin well  
that look at  
how the math  
works  
and why does  
the math  
work that  
way because  
of the  
general  
claim  
is that  
an argument  
what is  
the justification  
and the links  
among these  
dialectical  
categories  
well if there's  
three kinds of  
claims  
mathematical  
empirical  
and general  
then there  
are all the  
edges of that  
triangle  
if you had a  
fourth one  
you'd have all  
the edges  
of a tetrahedra  
and then you'd  
almost have a  
model

the mathematical  
to empirical  
direction  
invites  
philosophical  
analysis  
due to  
novel  
interpretations  
of scientific  
model terms  
so how can  
math be used  
to justify  
empirical claims  
why does  
equations  
have anything  
to do  
with what  
people say  
about the  
brain  
and the  
body  
and the  
niche  
why is that  
justified  
at all  
other than  
just other  
people in  
their epistemic  
authority  
have done  
it or it's  
how it has

been done  
Stephen  
instrumentalism  
also  
it gives  
a kind  
of a  
bridge  
between  
realist  
science  
and applied  
science  
and it's  
kind of  
sitting  
somewhere  
between  
the two  
so I  
think  
that's  
quite a  
useful  
piece  
that this  
adds  
to  
the  
equation  
nice  
and then  
the last  
two  
edges  
of that  
triangle  
are

how can  
mathematical  
not both  
directions  
on each  
edge  
you could  
add  
but how  
can  
mathematical  
claims  
justify  
general  
claims  
well how  
could you  
say that  
about  
nested  
systems  
about  
nested  
markov  
blankets  
the  
math  
okay  
that's  
important  
to  
investigate  
and how  
can  
general  
claims  
justify  
empirical

claims  
how  
could you  
say that  
about  
the  
ants  
well  
fitness  
or  
dissipative  
systems  
so how  
can  
general  
systems  
claims  
justify  
anything  
about  
empirics  
so those  
are some  
of the  
edges  
of these  
three  
kinds  
of  
claims  
and  
there's  
also  
claims  
and  
sub  
claims  
which



is  
why  
it's  
really  
important  
to have  
rhetorical  
ecosystem  
mapping  
and an  
ontology  
for active  
inference  
so that  
we can  
actually  
learn and  
apply  
across  
languages  
and through  
time  
Stephen  
yeah  
and I  
think  
also  
we can  
think  
of  
this  
as  
structural  
and  
functional  
insights  
being  
gained

rather  
than  
necessary  
repeatable  
measured  
outcomes  
every time  
being  
gained  
so  
there's  
an idea  
that  
these  
insights  
might  
reveal  
something  
about  
the  
structure  
or  
the  
functionality  
of  
the  
dynamics  
knowing  
that  
no  
one  
simulation  
may  
ever  
be  
repeated  
quite  
the

same  
way  
yes  
so  
the  
concluding  
remarks  
the  
active  
inference  
framework  
is  
incredibly  
ambitious  
is it  
the  
framework  
or the  
people  
in its  
explanatory  
scope  
from  
humble  
beginnings  
as a  
theory  
of  
brain  
function  
it  
is  
now  
positioned  
as a  
framework  
for  
understanding

life  
itself  
there  
is a  
critical  
tradition  
in  
the  
philosophy  
of  
biology  
inspired  
by  
Richard  
Levins  
with  
regard  
to  
such  
ambitions  
many  
then  
will  
approach  
active  
inference  
with  
skepticism  
healthy  
skepticism  
is a  
good  
thing  
but  
healthy  
skepticism  
is  
informed

skepticism  
unfortunately  
getting  
one's  
head  
around  
the  
details  
of  
active  
inference  
is  
no  
small  
task  
our  
goal  
in  
this  
introduction  
has  
been  
to  
clarify  
the  
basic  
mathematics  
history  
and  
internal  
dialectics  
of  
active  
inference  
in  
that  
order  
those

were  
the  
sections  
and  
draw  
attention  
to  
some  
key  
concerns  
with  
these  
details  
on the  
table  
philosophers  
of  
biology  
are  
in  
a  
better  
position  
to  
critically  
evaluate  
the  
framework  
we look  
forward  
with  
interest  
to  
seeing  
the  
results  
classic  
future

looking  
last  
sentence  
so  
that's  
the  
final  
lines  
we  
have  
a  
ton  
of  
things  
to  
discuss  
in  
37.1  
and 2  
it'd be  
awesome  
to have  
any  
of the  
authors  
as well  
as any  
other  
people  
who just  
want to  
jump on  
for a  
returning  
or a  
first time  
discussion  
but we'll

close  
under the  
buzzer  
in the  
third  
period  
I know  
that's  
how they  
count up  
there  
Dean  
so  
any  
final  
comments  
on  
just  
what you're  
looking forward  
to  
discussing  
Dean  
first  
because you  
never get  
the last  
word  
at the  
end  
Steven  
yeah  
I don't  
want the  
last  
word  
so  
one of



the  
things  
I want  
to talk  
about  
in the  
point  
one  
is  
the  
general  
claims  
because  
I get  
to trot  
out one  
of my  
favorite  
parables  
that I  
made up  
which is  
three  
functions  
walk  
into  
a  
bar  
and I  
want  
to be  
able  
to  
because  
I  
get  
lots  
of

mileage  
out  
of  
that  
and  
whether  
it's  
three  
functions  
or  
probability  
preference  
and  
fit  
the  
same  
holds  
true  
it  
can  
be  
great  
taste  
and  
less  
filling  
so  
I  
want  
to  
have  
a  
look  
at  
that  
because  
I  
think

that  
really  
tags  
onto  
the  
idea  
that  
what  
they're  
talking  
about  
here  
in  
terms  
of  
a  
guide  
could  
also  
set  
us  
up  
really  
well  
for  
creativity  
and  
then  
I  
also  
want  
to  
be  
able  
to  
say  
that  
that

creativity  
isn't  
because  
we're  
framed  
in  
but  
because  
we  
especially  
if we  
speak  
to  
the  
authors  
these  
first  
principles  
of  
hidden  
states  
and  
inferences  
and  
which  
ones  
are  
selected  
and  
which  
data  
is  
available  
to  
actually  
make  
inferences  
on

can  
be  
used  
particularly  
for me  
the example  
of  
psychology  
versus  
coaching  
psychology  
and  
that  
type  
of  
paradigm  
shift  
awesome  
yeah  
I really  
appreciated  
both  
of your  
perspectives  
on this  
paper  
it really  
shows  
how  
different  
life  
experiences  
can be  
in  
conversation  
with  
some  
of

the  
technical  
details  
so  
thank  
you  
both  
and  
for  
everybody  
who's  
watching  
and  
participating  
see you  
around  
the  
lab  
see you  
in  
the  
upcoming  
weeks  
for  
.1  
.2  
peace  
thanks  
guys  
bye  
!

Thank you.